

Programmable digital systems:

System verilog { system verilog language

ASIC work
- V - C "

RISC-V

Revision Part:

In vhdl all these dependencies with gate.

1 pico second $\hookrightarrow 10^{-12}$

variable out, out2: std_logic;

:= { variable

out1 := a and b;

out2 := out1 xor c;

out3 := (not out1) and out2;

} Variable o luyun
xin aninda deyer
bekleme olmaz.

combprocess(s, x0, x1)

begin

y <= x0;

if (s = '1') then

y <= x1;

endif;

end process comb;

Every circuit design on vhd on a stable state.

while loop for simulation hocu bunu synthesizable diye sorarsa ypk
for loop for synthesizable
 $\hookrightarrow$ synthesizable
desi.

In code intended for synthesis, use the smallest possible integer values and always limit the range.

Shift instructor $\rightarrow$ software part

Generation (generate):

Instead of using for loop we use generation constructor we have to
for for the process but if we have some ideas speed up ab outside of the process

obligatory

FG: for i in 0 to size-1 generate

sum[i] <= a[i] xor b[i] xor c[i];

c[i+1] <= (a[i] and b[i]) or - - - -

end generate;

In generation else and elsif branches are not allowed!

Nested generation

Accelerate circuit outside the process
not combinational sequential one

Generation in advantage bak

If $c_i = 0$ ilk boya (carry) atarsak simplify etde edelim [Full adderda]

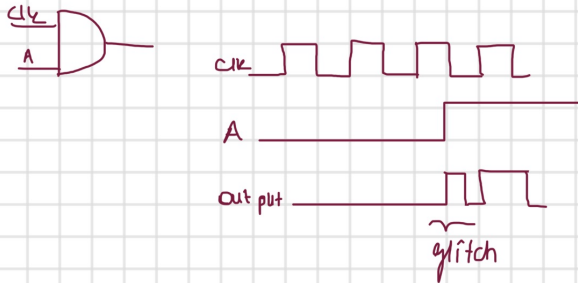
GUARDED other version of synchronous reset

clock has the higher priority and make asynchronous

D, T, JQ, ST
flip-flop



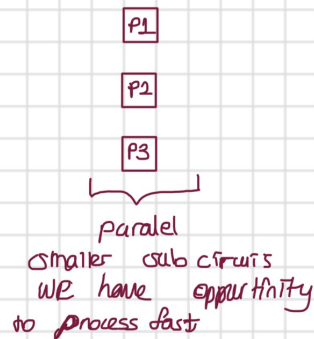
Clock gating



glitchler bazı durumlarda defru yalisabilir
ama bazka zamanlarda calismaya biler. Bundan
dolayi dikkatli olmak gerek

clock gating bunu
yapar.
her seyin clock ile senkronize
edilmesi daha saglikli

```
process (THREESTATE, CLK)
begin
  if (THREESTATE = "0") then
    OUTPUT <= "Z0";
  elsif (CLK'event and CLK = '1') then
    if (CONDITION = "1") then
      OUTPUT <= INPUT;
    end if;
  end if;
end process;
```



Wersja z nieprawidlowym opisen
yanlis versiyon

Design slayttan auzuk laka jelmeydi.
buna bak
Subtractor design part describe in vhdl

Sum $\Rightarrow$ Difference olmalı
 $C_i \Rightarrow$ Vif olmalı

Limiting the counting range:

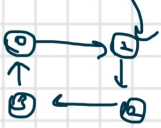
conv-integer:

→ integera dönüştürme

Karşılaştırma yapabilmek için

$\text{count_int} \leq X \text{ "00"}$; hexadecimal number demek

MOD 4



MOD N

